



Effective Fabrication and Characterization of Eco-friendly Nano Chitosan Capped Zinc Oxide Nanoparticles for Effective Marine Fouling Inhibition

Hend Al-aidy El-saied^a, Ahmed Mohammed Ibrahim^{b,*}

^a Egyptian Petroleum Research Institute, Petrochemicals Department, Polymers Lab, Nasr City, Cairo, Egypt

^b Marine Chemistry Lab., Marine Environment Division, National Institute of Oceanography and Fisheries (NIOF)-Suez Branch, Cairo, Egypt

ARTICLE INFO

Editor: Yunho Lee

Keywords:

Antifouling

Chitosan

Zinc oxide nanoparticles

Marine paint

Biofouling

ABSTRACT

Since the inhibition of organotin as antifouling additives in 2003, many scientists direct their efforts to develop novel environment-friendly marine antifouling additives. This work declares and evaluates the antifouling activity of Nano chitosan capped ZnO nanoparticles against primary biofilm and biofouling. Chitosan nanoparticles were prepared from the shrimp shell obtained, which will be environmentally friendly and will have no adverse effects on the environment's antialgal, antimicrobial and antifungal characteristics. Using co-precipitation method, the present study manages synthesis and characterization of Nano chitosan capped ZnO nanoparticles (NCCZO NPs) and was identified by FTIR, XRD, HRTEM, SEM, EDX, and UV-VIS. NCCZO NPs have been mixed in the prepared formula of inert marine paint. The formulated paints were then brushed twice on PVC panels, hung in a steel frame and immersed in the Mediterranean Sea Eastern Harbor of Alexandria, followed by visual examination and photographic recordings. The findings revealed once 64 days of immersion that (NCCZO NPs) are resistant to tubeworms and barnacles formation due to their functional groups. This inhibition activity was linked to the significant functional groups (hydroxyl and amine) of the NCCZO NPs. The strong antifouling activity makes them promising candidates for new antifouling additives.

1. Introduction

The primary cause of several environmental and health issue is bacterial biofilm. When a solid surface is submerged in a non-sanitary solution, a series of physical, chemical and biological occurrences are observed resulting in the creation of a complicated bio-community known as fouling [1]. Fouling is a significant issue in aquatic surfaces where marine microorganisms can bind to the surface and form a conditioning layer, providing a readily available platform for the attachment and proliferation of other aquatic organisms, such as diatoms and algae. This unwanted colonization has severe effects in the naval industry such as greater fuel consumption, loss of vessel maneuverability, greater corrosion rate, and blockage of seawater conducting installations' valves and tubes. Fouling organisms may also affect biomedical implants, surgical equipment, protective clothing in hospitals, are also impacted by the fouling of possibly device failure organisms. For example, fouling can trigger harmful biological processes such as platelet adhesion, aggregation, and foreign body reaction [2–4].

During the mid-1800s, a variety of paints were developed based on the idea of dispersing a toxicant in a polymer vehicle. In addition to multiple types of resins as binders, the oldest methods used materials

such as arsenic, mercury oxide, heavy metals (lead) and tri-butyl tin-based compounds as antifouling agents, such as turpentine oil, naphtha, benzene, linseed oil, and shellac varnish. Several studies have shown that these kinds of biocide compounds, owing to their elevated persistence and toxicity, have a negative impact on aquatic life and more specifically on non-target fouling species such as oysters [5–7]. In 2003, the International Maritime Organization (IMO) prohibited the use of these biocidal kinds in the production of antifouling paints [8]. Coatings for antifouling (AF) have been created to inhibit the development of biofilms to avoid surface bacterial adhesion, which is a critical stage during colonization. Compounds with quaternary ammonium functionality as well as poly (ethylene glycol) (PEG) and its derivatives can kill connected bacteria because of their elevated hydration forces that decrease protein and cell adherence. Recently, Zwitterionic structures have been discovered to be capable of stopping original, non-specific adhesion of microorganisms, containing both positive and negative charges within one molecule. Due to their low surface energy and good non-stick features, fluoropolymers were used as great fouling release coatings. Such copolymers have also produced polymer brushes and used them in multiple antifouling coatings [9–12]. Many materials with interesting characteristics have been generated by nanotechnology.

* Corresponding author.

E-mail addresses: hend_alaidy@yahoo.com (H.A.-a. El-saied), ahmed_pending@yahoo.com (A.M. Ibrahim).

<https://doi.org/10.1016/j.jece.2020.103949>

Received 24 November 2019; Received in revised form 11 April 2020; Accepted 13 April 2020

Available online 18 April 2020

2213-3437/ © 2020 Elsevier Ltd. All rights reserved.

Among the transition metal oxides, (e.g. Au, Ag, Cu, TiO₂, ZnO and CuO) have received the most attention due to their wide applications in enormous area of research [13–17]. Successive long-term procedure to avoid biofilm development needs stable and antifouling nanomaterial development. Therefore, controlling bacterial adhesion on nanomaterials by antimicrobial layer is essential for developing antifouling material for effective water therapy [18,19]. The costs connected with material synthesis, however, are much greater. It is therefore essential to develop an easy and expedient strategy for the synthesis of stable nanomaterials to avoid pore diffusion and the formation of biofilms. ZnO nanoparticles have brought attention in comparison with other metal oxide nanoparticles owing to their reduced price, absence of color and UV blocking characteristics, among others [20–22]. Nano ZnO can be synthesized in a wide variety of forms: rods, cables, whiskers, belts, bipods, tetrapods, tubes, flowers, propellers, bridges, and cages [23–25]. ZnO-NPs were prepared using methods such as sol-gel [26,27], precipitation [28], hydrothermal synthesis [29], and pyrolysis spray [30,31]. Chitosan is one of the most abundant biopolymers on Earth and includes (1,4)-linked 2-amino-2-deoxy-β-D-glucose cationic polysaccharide and 2-acetamido-2-β-D-glucose units [32,33]. It is a naturally occurring polymer that is fully biodegradable and biocompatible and can be used as adhesive, antibacterial and antifungal agent [34–36]. Chitosan's primary amine and hydroxyl groups have a very powerful affinity (like a chelating agent) to metal ion to decrease particle size and stop agglomeration [37]. Herein, we reported the preparation of Nano chitosan capped ZnO nano particles (NCCZO NPs) through co-precipitation technique in aqueous medium with eco-friendly green capping agent (Nano chitosan) under atmospheric condition at 80 °C. The resulting samples are characterized by studies involving spectral [Fourier-transform infrared spectroscopy (FT-IR) and X-Ray Diffraction (XRD)], microscopic [Field Emission Scanning Electron Microscope (FESEM) and High-resolution transmission electron microscopy (HRTEM)], and thermal [Thermogravimetric analysis (TGA)] characterization techniques. To test their antifouling activity against marine fouling organisms such as barnacles and tubeworms, these particles were placed in an inert marine paint formulation. NCCZO NPs show 90 % antifouling effectiveness against the development of marine fouling bacteria to be used as a promising antifoulant.

2. Experimental

2.1. Materials

Shrimp shell waste was acquired from the Egyptian local fish market, Sodium tripolyphosphate (TPP), acetic acid, sodium hydroxide, hydrochloric acid, KMnO₄, Oxalic acid and Zinc nitrate (hexahydrate) were obtained from the El-Nasr Company for Intermediate Chemicals, Egypt. All other solutions were prepared with Deionized water (DIW). We also purchased paint components such as oil binder, iron oxide, zinc oxide, kaolin and xylene from Company of Chemical Partners, Egypt. All chemicals utilized as received were of laboratory grade.

2.2. Preparation of Chitosan from shrimp shell

As earlier stated in the procedure [38], chitosan extraction from shrimp shells was performed. Local shrimp waste was collected and instantly washed several times with water and detergent, then permitted to dry out in the open region under sunlight and then ground into small particles. The grounded material was converted into chitosan by three consecutive steps (demineralization, deproteinization, and deacetylation process (DA) to remove acetyl groups [39]. The shrimp shells that were purchased were then soaked in 1 M NaOH for 24 hrs, washed and dried. Then 1 M HCl was used to demineralize the powder of the shrimp shells. By decomposing calcium carbonate into water-soluble calcium salts, demineralization was easily achieved with the release of carbon dioxide. Then the solution was filtered and the residue

was washed with deionized water (DI) several times until the pH was neutral. Discolored with KMnO₄ and oxalic acid to acquire chitin powder during the DA stage, the acetyl groups were segregated to convert chitin to chitosan. The chitin generated was then deacetylated using 50 percent NaOH, which was repeated to achieve a higher degree of chitosan deacetylation [40–42]. Alkali solution was drained and washed with DI constantly until pH was reduced to normal. Finally, chitosan was dried and stored for further use at room temperature.

2.3. Preparation of chitosan nanoparticles (ChNPs)

ChNPs were prepared by dissolving in 250 ml of 1 percent acetic acid (v / v) at 2.5 g of prepared chitosan and stirred with a magnetic at room temperature overnight. The solution's pH was adjusted utilizing 1 M NaOH to be 4.8. At a concentration of 0.8 μg / mL, tripolyphosphate (TPP) was dissolved in DI water. TPP solution has been introduced to the chitosan solution at distinct original ratios of chitosan to TPP and chitosan nanoparticles were spontaneously shaped at room temperature under vigorous magnetic stirring. Nanoparticles were divided at 10,000 rpm for 30 min by centrifugation [43]. The resulting nanoparticles of Chitosan were cleaned with DI water five times to remove any residues of TPP, NaOH. Finally, before further use or analysis, the pellet acquired was dried at 40 °C for 24 hours [44,45].

2.4. Preparation of uncapped ZnO NPs (UCZO NPs) and Nano chitosan capped ZnO nano particles (NCCZO NPs)

Uncapped ZnONPs (UCZONPs) and Nano chitosan capped ZnO nanoparticles (NCCZO NPs) were synthesized using chemical precipitation method using zinc nitrate and NaOH as precursors in presence of soluble starch as stabilizing agent. In brief, 1.0 g of ChNPs dissolved in 40 mL H₂O and 1.0 ml (5%) of acetic acid, the solution was stirred for 30 min and then 25 mL of zinc nitrate hexa hydrate (0.25 M) to a solution 0.85% starch was added to the above solution and continuously stirred for 10 minutes to obtain a homogeneous solution and dissolved completely and stirring is continued for 2 h at 80 °C. With this solution, 50 mL of sodium hydroxide (1 M) solution was added drop by drop till the precipitate was formed. The formed white precipitate was allowed to settle for 24 h by centrifugation at 3000 rpm and the precipitate was washed with DI water and ethanol five times. NCCZO NPs were obtained and dried at 80 °C overnight. The same procedure for uncapped ZnONPs (UCZONPs) is followed as mention above but in absence of ChNPs.

2.5. Materials characterizations

Characterization of chitosan from shrimp shells by (i) Ash value as 2 g of chitosan sample is placed in a clean, dry, pre-weighted porcelain crucible and heated in a 5-h muffle furnace preheated to 700 °C. Ash value percent is calculated using the following [46].

$$\text{Ash\%} = \frac{W_1 - W_2}{W_1} \times 100$$

Where W₁ and W₂ in grams are the weight of chitosan before and after ignition, respectively.

(ii) Degree of DA by the acid-base titration technique with changes as (0.4 g) chitosan was dissolved in 60 ml HCl (0.1 mol/l) at room temperature with 3-6 drops of methyl orange and then the red chitosan solution was titrated with 0.1 mol/l NaOH solution until it turned orange.

The DA was calculated by the formula

$$\text{DA\%} = \frac{C_1 V_1 \times C_2 V_2}{M \times 0.0994} \times 0.016$$

Where C₁ = concentration of standard HCl aqueous solution (mol/l), C₂ = standard NaOH solution (mol/l), V₁ = volume of the standard HCl

aqueous solution for dissolving chitosan (ml), V_2 = volume of standard NaOH solution consumed during titration (ml), and M = weight of chitosan (g). The number 0.016 (g) is the corresponding NH_2 group weight in 1 ml of standard 1 mol/l HCl aqueous solution and 0.0994 is the percentage of NH_2 group by chitosan weight [47].

The chitosan (Ch), chitosan nanoparticles (ChNPs), uncapped ZnO NPs (UCZO NPs) and Nano chitosan capped ZnO nanoparticles (NCCZO NPs) were characterized by following methods: "FT-IR spectroscopy was performed in a Shimadzu-FTIR 4200 spectrometer and the spectra were recorded in the frequency range of 400 to 4000 cm^{-1} using KBr as reference". A modern PAN analytical diffractometer, the Xpert PRO model, was used to perform XRD. Nickel filtered copper radiation was used ($\lambda = 1.542 \text{ \AA}$). At room temperature and under constant operating conditions (40 KV & 40 mA), all the diffraction patterns were examined. The samples were registered at diffraction angle (2θ) range between 5° and 80° . The absorption spectrum was researched using a 200-800 nm wavelength variety of UV-Vis spectrophotometer (UV-2450, Shimadzu, Japan). An HR-SEM (JEOL, JSM model no: 6360) equipped with an EDX elemental composition analyzer was used to analyze the surface morphology and elemental analysis of the samples and working at an accelerating voltage of 200 KV. The dried samples in the liquid nitrogen were freeze-fractured and then coated with platinum before observation and photography. TEM -1230 with an accelerating voltage of 100 KV (JEOL Co., Japan) was used to examine size, shape, and morphologies of the nanoparticles. After sonication, the samples were prepared on a copper grid.

2.6. Antifouling test

2.6.1. Paint preparation

The formulation of marine paint consisted of 20 g zinc oxide, 40 g of oil binder, 20 g kaolin, 5 mL xylene, and 15 g of iron oxide. These components mixed together in porcelain bottle jar and some porcelain balls were added for stirring on stone ware instrument to form homogeneous paint. Control panel was prepared by this formulation without any additive. We took 2 g from Nano-chitosan (NC) and NCCZnO NPs as additives and mixed with 48 g from marine paint formulation separately and applied on NC and NCCZnO NPs panels [48].

2.6.2. Panel and frame preparation

A PVC panel (0.2 cm) thick sheet was sliced into $10 \times 15 \times 0.2 \text{ cm}^3$ sheets that were roughened using emery papers starting with a coarse one and proceeding in steps to finer grades. Two consecutive coats of the formulated paint were covered on both sides of these panels. The paints were made by mixing certain quantities of binder, pigments, plasticizer, then NC, NCCZnO NPs and high-speed solvents centrifuging ball mill, were allowed to dry for two days between each coating. The covered panels were linked by nails bored in the panels to the testing iron frames with nylon threads [49].

2.6.3. Field anti-macrofouling assays at Eastern Harbor of Alexandria beach (Mediterranean Sea)

All panels (uncoated, control, NC and NCCZnO NPs) were immersed at a depth of 1.5 meters in the Eastern Harbor of Alexandria (Fig. 1), where visual inspection and photographic recording periodically studied the antifouling performance of each coated panel from 9 December 2018 to 12 February 2019. The models were drawn out of the ocean after a certain moment and cleaned thoroughly with seawater and photographed. They were then put in the seawater instantly to continue the experiment. The coverage percentage of marine fouling organisms was used as parameters for expressing macrofouling propensity over different time intervals.

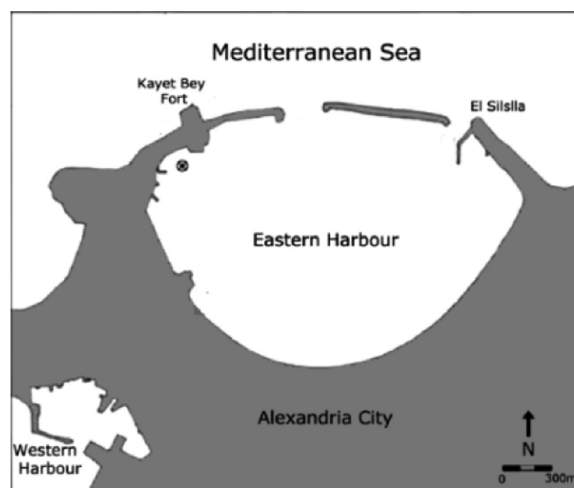


Fig. 1. Marine location of panel immersion test in front of NIOF, E. H. of Alexandria, Egypt.

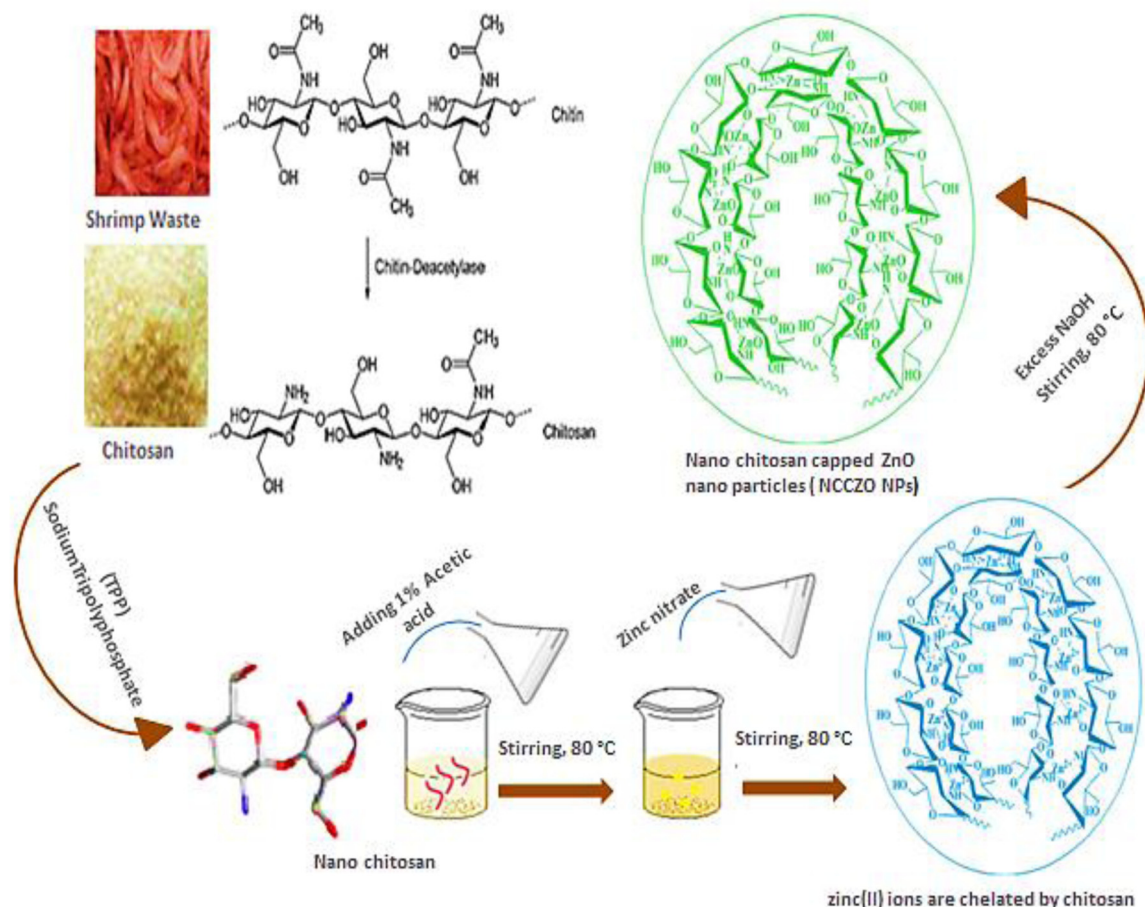
3. Results and Discussion

3.1. Mechanism for the formation of (NCCZO NPs)

Scheme 1 showed the synthesis of the Nano chitosan capped ZnO nanoparticles (NCCZO NPs). During the DA process with NaOH, the extracted chitosan from shrimp waste was obtained at low M.wt (3923 D) by continuous heating of the polymer causing a decrease in the M.wt with the increase in the DA degree [50]. It was discovered that the smallest M.wt was selected to be simpler to resize in the nano form, where 260 nm was generated. Chitosan's degree of DA and ash content was discovered to be 68% and 9% respectively. The DA is defined as the number of amines group in relation to the number of amides group of the polymeric chain formed as a result of the removal of $(-\text{CO}-\text{CH}_3)$ group from the chitin chain as shown in Scheme 1. It is the most widely used parameter for characterizing chitosan. In the absence and presence of Nano chitosan solution, both uncapped ZnO NPs and Nano chitosan capped ZnO nanoparticles were produced using the co-precipitation method using zinc oxide and NaOH as precursors. The synthesis technique of co-precipitation (NCCZO NPs) and (UCZO NPs) has a growth benefit and the manufacturing process is cheap and eco-friendly. In the synthesis of controllable nanostructure size and morphology, various variables such as pH, temperature, capping agent, modulator reagent and reaction time play a significant part. Zinc (II) ions are chelated by nano chitosan biopolymer and subsequently, interact with sodium hydroxide resulting in colloidal zinc hydroxide solution being transformed to zinc oxide nanoparticles. Since they aggregate instantly, the synthesis of uncapped zinc oxide NPs in aqueous solution at low temperature and without capping agent was not simple and some impurities such as $\text{Zn}(\text{OH})_2$ are also present. Acetic acid was used as a modulator reagent to prevent the interaction between the metal ions and the organic linkers, creating a competitive situation that regulates the rate of frame extension and the growth of crystals. Few drops of acetate acid have been added to boost chitosan solubility that is obtained from chitin and has a group of acetates. In the presence of acetic acid and excess sodium hydroxide solution, the zinc hydroxide was completely converted to zinc oxide nanoparticles without any impurity such as zinc hydroxide or starting material and stirring vigorously at 80°C .

3.2. FTIR spectroscopy

Fig. 2 displays (A) Chitosan, (B) Chitosan Nanoparticles (ChNps), (C) Uncapped ZnO NPs (UCZO NPs) and (D) Nano Chitosan Capped ZnO Nanoparticles (NCCZO NPs) FT-IR spectra. Fig. 2(A) shows the FT-IR spectrum of chitosan extracted from chitin, where the prepared



Scheme 1. The scheme showing the synthesis of Nano chitosan capped ZnO nano particles (NCCZO NPs).

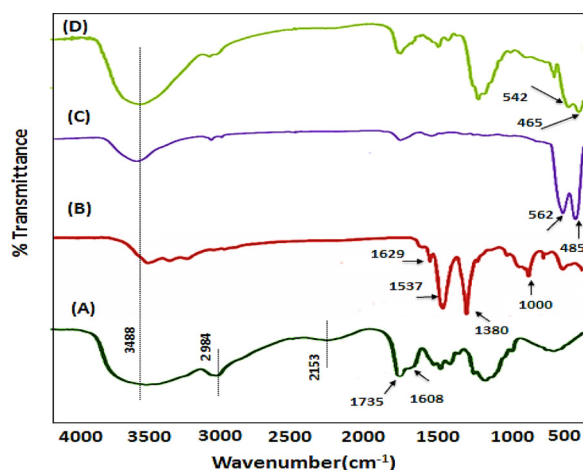


Fig. 2. FT-IR spectra of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) uncapped ZnONPs (UCZO NPs) and (D) Nano chitosan capped ZnO nano particles (NCCZO NPs).

chitosan spectrum shows broader absorption bands at 3488 cm⁻¹ than those found in the Chitosan Nanoparticles (ChNPs) spectrum Fig. 2(B) which indicating that the hydrogen bonding is enhanced. This peak is attributed to -NH₂ and -OH groups stretching vibration. The frequencies at 2984 and 2153 cm⁻¹ correspond to aliphatic -C-H symmetric and asymmetric stretching vibrations respectively. The peaks at 1735 cm⁻¹ and 1608 cm⁻¹ are attributed to the CONH₂ and NH₂ groups, respectively. These peaks shift hypsochromically to 1629 and 1537 cm⁻¹ in the

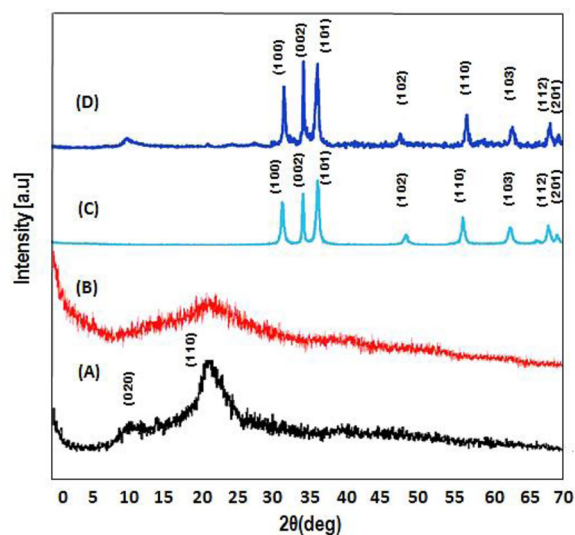


Fig. 3. XRD pattern of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) uncapped ZnONPs (UCZO NPs) and (D) Nano chitosan capped ZnO nano particles (NCCZO NPs).

FTIR spectra of (ChNPs) which caused by the interaction between NH₃⁺ groups of chitosan and phosphate groups of TPP. This interaction is also supported by the decrease of amide band I (1629 cm⁻¹) intensity in (ChNPs) compared to chitosan. The other significant band for (ChNPs) was observed at 1380 cm⁻¹ owing to -CH₂ wagging. Furthermore, the

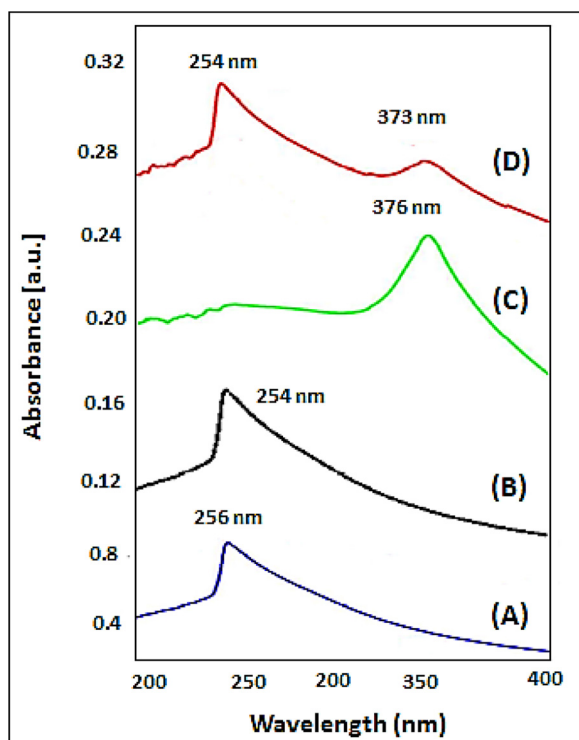


Fig. 4. UV-Visible spectra of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) uncapped ZnONPs (UCZO NPs) and (D) Nano chitosan capped ZnO nano particles (NCCZO NPs).

peak at 1000 cm^{-1} which appears in the FTIR spectra of (ChNPs) shows characteristic of $\text{P}=\text{O}$ stretching vibration from phosphate groups [51–53]. The IR spectrum of uncapped ZnO NPs (UCZO NPs) Fig. 2 (C) usually shows a characteristic absorption band between 485 and 562 cm^{-1} due to two transverse optical stretching modes of ZnO [54].

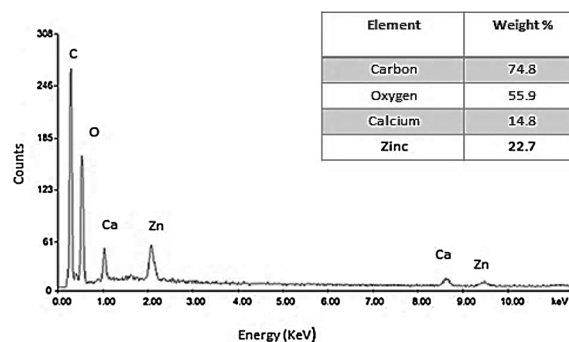


Fig. 6. EDX spectrum of Nano chitosan capped ZnO nano particles (NCCZO NPs).

The peak at 3488 cm^{-1} corresponding to the stretching vibration of $-\text{OH}$ and $-\text{NH}_2$ became broader and of higher intensity indicating some interaction between these groups and ZnO NPs Fig. 2(D) [55]. Compared with pure chitosan, new bands appearing at 465 and 542 cm^{-1} can be attributed to the stretching vibration of $\text{N}-\text{Zn}-\text{O}$ and $\text{O}-\text{Zn}-\text{O}$ [56].

3.3. XRD analysis

The Fig. 3 shows XRD patterns of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) uncapped ZnO NPs (UCZO NPs) and (D) Nano chitosan capped ZnO nano particles (NCCZO NPs) with 2θ values in the range of 5° and 70° and the reflections are labelled with the Miller indices. From the XRD patterns of the extracted chitosan and chitosan nanoparticles Fig. 3 (A&B), after crosslinking with TPP during the formation of chitosan nanoparticles, the two characteristic peaks at 10° and 20° which correspond to (020), (110) planes of the extracted chitosan with crystalline structure were completely destroyed, where the peak at 10° vanished in the ChNPs pattern indicating the amorphous nature, this in agreement with a previous work [57] The decreasing of the polymer crystallinity leads to an enhancement in the formation of

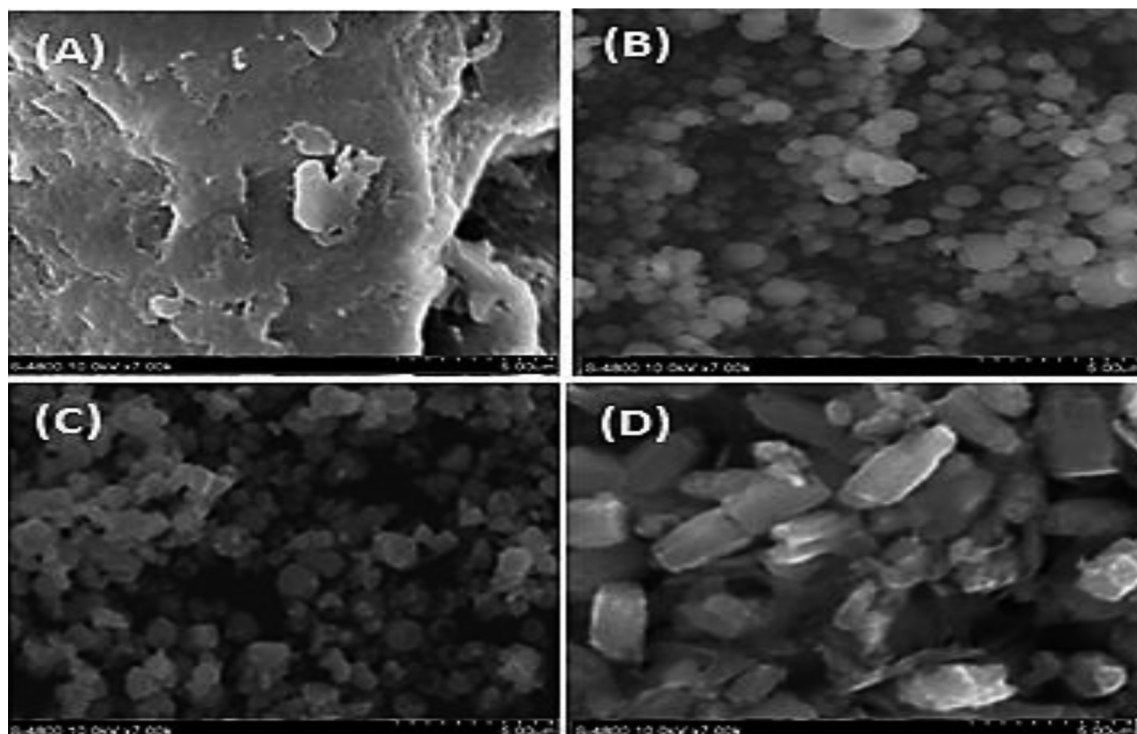


Fig. 5. SEM images of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) uncapped ZnO NPs (UCZO NPs) and (D) Nano chitosan capped ZnO nano particles (NCCZO NPs).

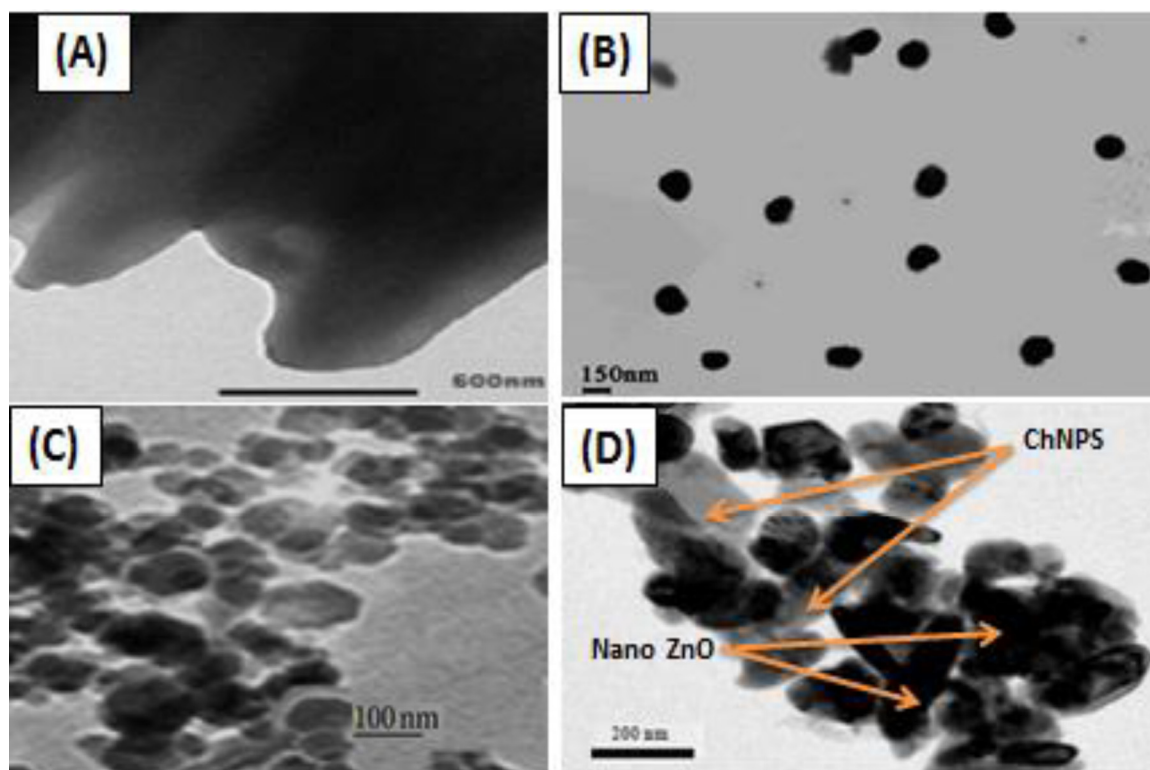


Fig. 7. TEM images of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) uncapped ZnONPs (UCZO NPs) and (D) Nano chitosan capped ZnO nano particles (NCCZO NPs).

NCCZO NPs. The main characteristic peaks for (C) uncapped ZnO NPs (UCZO NPs) were observed at 31.69° , 34.35° , 36.30° , 47.55° , 56.58° , 62.80° , 68.04° and 69.60° which correspond to (100), (002), (101), (102), (110), (103), (112) and (201) planes. The peaks verify the formation of pure uncapped ZnO NPs (UCZO NPs) without impurity. We can only obtain a definite structure and nanosized shape in the presence of a capping agent (ChNPs), which is verified by SEM and TEM data analysis. Even though, the above findings indicate that without capping agent (ChNPs) we can synthesize pure ZnO, but it does not form in a definite framework and nanosized form. Fig. 3 (D), with the interruption of intermolecular hydrogen bonding within the ChNPs, the intensity of diffraction peak at $2\theta = 10^\circ$ and 20° corresponding to Chitosan became so weak. Thus, the molecular chain of Chitosan was less stacked, forming the NCCZO NPs complex. The main characteristic peaks for zinc oxide were observed for the NCCZO NPs complex indicating uniform dispersion of ZnO NPs in (ChNPs) network Fig. 3(D). The average crystallite particle size of NCCZO NPs calculated to the most intense peak at 47.55° using Scherrer formula is 50 nm.

3.4. UV-Vis analysis

UV-Visible spectroscopy investigates the optical characteristics of (A) chitosan, (B) chitosan nanoparticles (ChNPs), (C) uncapped ZnO NPs (UCZO NPs) and (D) nano chitosan capped ZnO nano particles (NCCZO NPs). Fig. 4 (A&B) demonstrates the chitosan and ChNP absorbance range showing the maximum absorbance at 256 nm, respectively, 254 nm. A maximum peak was observed at 376 nm, corresponding to the distinctive absorption peak for uncapped ZnO NPs (UCZO NPs) Fig. 3 (C). The Nano chitosan capped ZnO nanoparticles (NCCZO NPs) showed an absorption band at 254 nm and 373 nm, confirming the existence of ZnO NPs in the ChNPs networks Fig. 4 (D). This shows that there were more uniform and dispersed ZnO particles in nano chitosan capped ZnO nanoparticles (NCCZO NPs). The encapsulation of chitosan is thought to limit the agglomeration of

particles and thus the growth of ZnO. In SEM photograph Fig. 4(D) of NCCZO NPs, uniform distribution of nano agglomerated ZnO is obviously shown compared to UCZO NPs Fig. 4(C).

3.5. SEM and EDX analysis

SEM analysis explored the morphology of (A) chitosan, (B) chitosan nanoparticles (ChNPs), (C) uncapped ZnO NPs (UCZO NPs) and (D) nano chitosan capped nanoparticles (NCCZO NPs). As with regular granular morphology, the surface of Chitosan has sheet form Fig. 5 (A), and Chitosan Nanoparticles (ChNPs) has either oval or circular shape Fig. 5 (B). Fig. 5 (C) displays an image of synthesized UCZO NPs that demonstrates obviously that the zinc oxide has aggregated featureless morphology and sort of rod. This sort of aggregation was primarily due to the elevated surface energy of UCZO NPs [58,59] which resulted in the creation of reduced reactivity bulk structures. On the contrary, compared to UCZO NPs Fig. 5(D), NCCZO NPs were found to be more distinct and uniform. In the Nano chitosan matrix, when ZnO nanoparticles were incorporated into nano chitosan, the surface is observed in rod form. It is concluded that the formation of rod shaped zinc oxide is possible only in the presence of Nano chitosan. EDX analyzed the elementary structure of NCCZO NPs. Fig. 6 illustrates that there are only Zn, C, O and Ca in the NCCZO NPs and there are no other impurities. The analysis proposed the successful formation of ZnO nanoparticles and the homogeneous dispersion within the matrix of ChNPs.

3.6. TEM analysis

TEM evaluated the shape and size of (A) Chitosan, (B) Chitosan Nanoparticles (ChNPs), (C) Uncapped ZnO NPs (UCZO NPs) and (D) Nano Capped ZnO Nano Particles (NCCZO NPs). Fig. 7 (A) displays a chitosan TEM image that obviously demonstrates the chitosan's sheet shape. Chitosan nanoparticles (ChNPs) are almost spherical in form, smooth surface and size range of approximately 260-150 nm Fig. 7 (B).

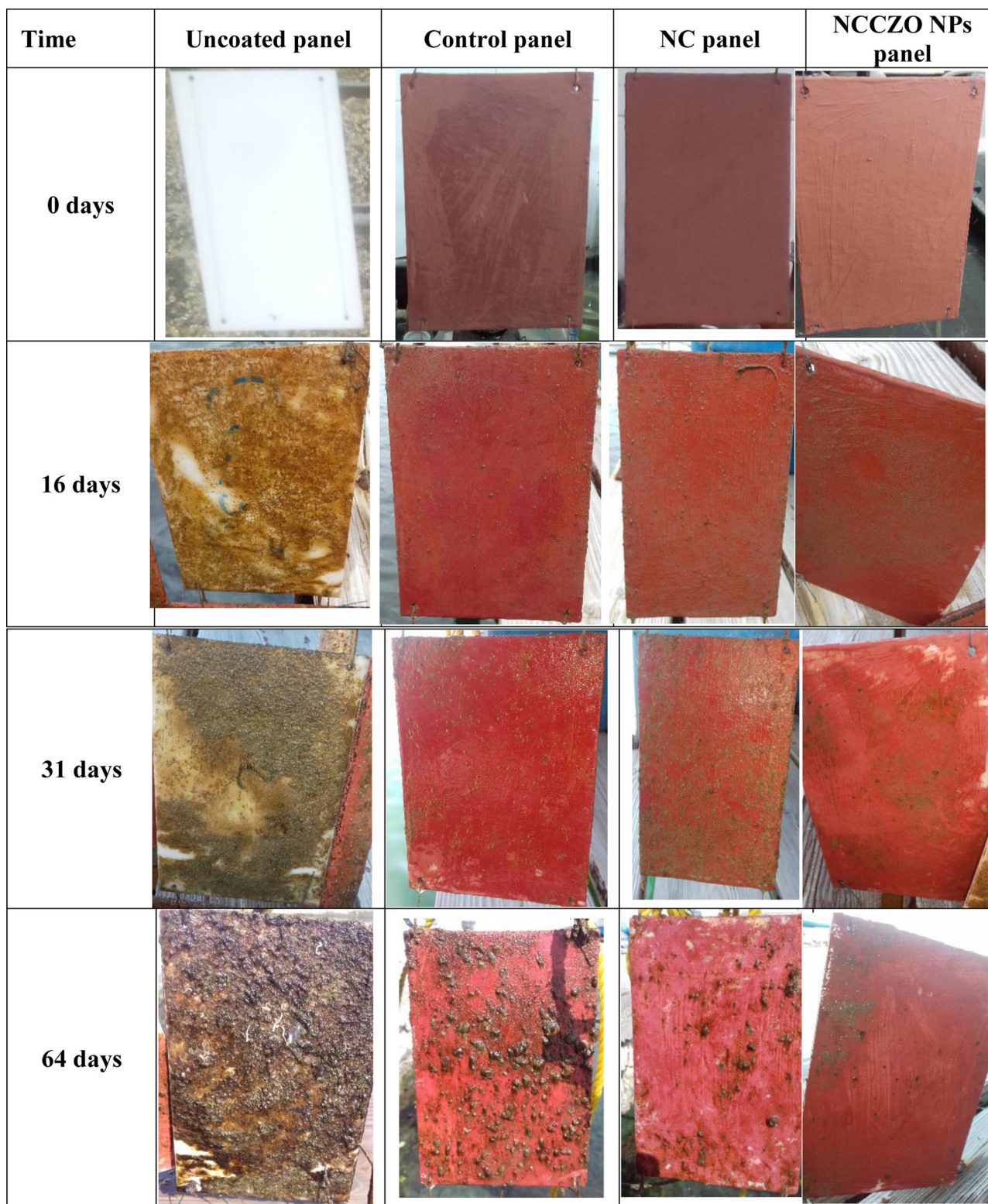


Fig. 8. Photographs collected from field test of anti-macrobiofouling assay for NCCZO NPs panel (Paint formulation based on NCCZO NPs), NC panel (paint formulation based on NC) in comparison to control (Paint formulation without any additives) and uncoated panel (blank) (Panel without any paint formulation).

UCZO NPs only displays aggregated shape, some of them are irregular in size, which is confirmed in Fig. 7 (C). Fig. 7 (D) reveals the NCCZO NPs TEM image and shows the dark rod shape parts surrounded by white features that correspond to UCZO NPs symmetrically capped by Nano chitosan biopolymer. It was expected that the mean particle size would be around 70-50 nm.

3.7. Antifouling test

During the period from 9 December 2018 to 12 February 2019, the antifouling test was applied for uncoated, control, NC, NCCZO NPs PVC panels and hanged on a steel frame and immersed in the Eastern Harbor of Alexandria, Egypt at a depth of 1.5 m. The uncoated panel was

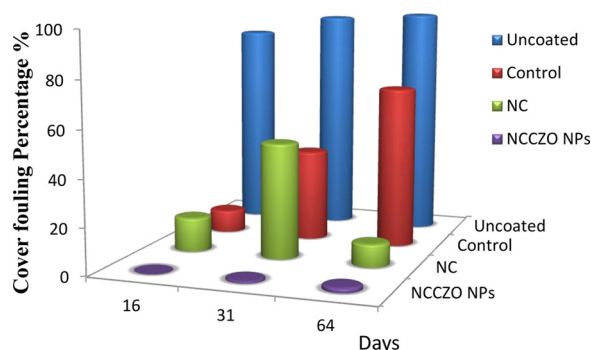


Fig. 9. Cover fouling percentage with time intervals for blank, control, NC and NCCZO NPs PVC panels immersed in Alexandria Eastern Harbor, Egypt starting from December 2018 to February 2019.

coated with about 90% mucous slime film (marine bacteria and green algae) after 16 days of immersion, while the control panel contained slime film with 10% on its sides and NCCZO NPs panel was free of slime film (Fig. 8). Due to some boats' motion, some silt spots existed on NC and NCCZO NPs panels. After 31 days of immersion, some tubeworms appeared on the surface of the uncoated panel, covered with slime film. The control panel contained biofilm, spread over its entire surface by 40%. NC panel contained biofilm and silt layer with 50%. The NCCZO NPs panel was still free of biofilm, and tubeworms appeared on its surface except for a small quantity of silt (Fig. 8). The uncoated panel was completely fouled with tubeworms, barnacles, and algae after 64 days of immersion. The control panel included 70 percent barnacles and tubeworms. The formed layer on NC panel was fallen and contained fouling with 10% due to the dissolution of chitosan nanoparticles. While, the NCCZO NPs panel was resistant to tubeworms, barnacles and algae which generated only on one side of the edges of its surface owing to the release of NCCZO NPs into seawater (Fig. 8). Fig. 9 displays the percentage of formation of different marine fouling organisms at different time intervals for blank, control and NCCZO NPs panels.

4. Conclusions

The current work directed at synthesizing and characterizing Nano chitosan and Nano chitosan capped ZnO nanoparticles (NCCZO NPs) using co-precipitation techniques and illustrating their structures through FTIR, XRD, UV-VIS, SEM, EDX, and HRTEM. The NC and (NCCZO NPs) mixed with the prepared formulation of inert marine paint. The resulting paints were then brushed twice on PVC panels, hanged in a steel structure and immersed in Eastern Harbor of Alexandria Mediterranean Sea followed by visual inspection and photographic recordings. The findings indicated that NC and (NCCZO NPs) inhibit the development of tubeworms and barnacles owing to their functional groups after more than two months of immersion. The efficiency of NCCZO NPs was larger than NC in inhibition of biofouling. Synergic effect between chitosan nanoparticles (ChNPs) & ZnONPs was considered the reason for this inhibition. As NCCZO NPs could attach to the barnacles and tubeworms cell membrane through $-NH_2$ group of ChNPs and enhanced the cell membrane permeability leading to cell destruction. There was also a slow release of Zn^{2+} from the NCCZO NPs in the reaction media which may have enhanced the inhibitory effect of the NCCZO NPs. So, NCCZO NPs can be considered promising anti-fouling. This also paves the way for an environmentally friendly anti-fouling coating to be developed.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

Acknowledgments

The authors are thankful to the Egyptian Petroleum Research Institute for promoting all the problems we faced through our research and offering all the research completion possibilities. Also, we would like to thank the National Institute of Oceanography and Fisheries (NIOF), Egypt for his economic assistance for this research.

References

- [1] M. Lejars, A. Margaillan, C. Bressy, Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings, *Chem. Rev.* 112 (2012) 4347–4390.
- [2] D. Rana, T. Matsuura, Surface modifications for antifouling membranes, *Chem. Rev.* 110 (2010) 2448–2471.
- [3] G. Mazue, R. Viennet, J.Y. Hihn, L. Carpentier, P. Devidal, I. Albaina, Large-scale ultrasonic cleaning system: design of a multi-transducer device for boat cleaning (20 kHz), *Ultrason. Sonochem.* 18 (2011) 895–900.
- [4] A.M. Brzozowska, F.J. Parra-Velandia, R. Quintana, Z. Xiaoying, S.S.C. Lee, L. Chin-Sing, D. Jan'czewski, S.L.M. Teo, J.G. Vancso, Biomimicking micropatterned surfaces and their effect on marine biofouling, *Langmuir* 30 (2014) 9165–9175.
- [5] M. Edge, N.S. Allen, D. Turner, J. Robinson, S. Ken, The enhanced performance of biocidal additives in paints and coatings, *Prog. Org. Coat.* 43 (2001) 10–17.
- [6] J. Thomas, S. Choi, R. Fjeldheim, P. Boudjouk, Silicones containing pendant biocides for antifouling coatings, *Biofouling* 20 (2004) 227–236.
- [7] A.G. Nurioglu, A.C.C. Esteves, G. d. With, Non-toxic, non-biocide-release antifouling coatings based on molecular structure design for marine applications, *J. Mater. Chem., B* 3 (2015) 6547–6570.
- [8] I. Banerjee, R.C. Pangule, R.S. Kane, Antifouling coatings: recent developments in the design of surfaces that prevent fouling by proteins, bacteria, and marine organisms, *Adv. Mater.* 23 (2011) 690–718.
- [9] W. Zhao, Q. Ye, H. Hu, X. Wang, F. Zhou, Grafting zwitter ionic polymer brushes via electrochemical surface-initiated atomic-transfer radical polymerization for antifouling applications, *J. Mater. Chem., B* 2 (2014) 5352–5357.
- [10] J. Yao, S. Chen, C. Ma, G. Zhang, Marine anti-biofouling system with poly-(3-caprolactone)/clay composite as carrier of organic antifoulant, *J. Mater. Chem., B* 2 (2014) 5100–5106.
- [11] B. Lallo da Silva, B.L. Caetano, B.G. Chiari-Andréo, R.C.L.R. Pietro, L.A. Chiavacci, Increased antibacterial activity of ZnO nanoparticles: Influence of size and surface modification, *Colloid. Surface. B* 177 (2019) 440–447, <https://doi.org/10.1016/j.colsurfb.2019.02.013>.
- [12] R.M. Fikry, R.A. El-Adly, N.A. Ismail, A.S. El-Tabei, H. Al-Aidy, Some azine and azole derivatives as antioxidant additives for lithium lubricating grease, *Egypt. J. Pet.* (2013).
- [13] J. Kreuter, Nanoparticles- a historical perspective, *Int. J. Pharm.* 331 (1) (2007) 1–10.
- [14] (a) D. Ali, M. Sasikala, M. Gunasekaran, N. Thajuddin, Biosynthesis and characterization of silver nanoparticles using marine cyanobacterium *Oscillatoria willei* NTDM01, *Dig. J. NanomaterBios* 6 (2011) 385–390; (b) M. Rai, A. Yadav, A. Gade, Current trends in phytosynthesis of metal nanoparticles, *Crit. Rev. Biotechnol.* 28 (2008) 277–284.
- [15] I. Freestone, N. Meeks, M. Sax, C. Higgitt, 'The Lycurgus Cup: A Roman nanotechnology', *Gold Bulletin* 4 (2007) 270–277.
- [16] Webmineral, Aeschnite-(Ce): mineral information and data, Retrieved, (2011) 10-08.
- [17] H.A. Raslan, H.A. El-Saied, R.M. Mohamed, N.M. Yousif, Gamma radiation induced fabrication of styrene butadiene rubber/magnetite nanocomposites for positive temperature coefficient thermistors application, *Compos. Part B Eng.* 176 (2019) 107326, <https://doi.org/10.1016/j.compositesb.2019.107326>.
- [18] T. Robert, C. Sylvain, O. Todd, P. Patrick, G. Andrey, L. Wei, R. Gary, P. Ravi, T. Himanshu, Small particles, Big impacts: A review of the diverse applications of Nanofluids, *J. Appl. Phys.* 113 (2013) 011301(b) en. wikipedia.org.
- [19] (a) P.N. Ezhilarasi, P. Karthik, N. Chhanwal, Relevance and challenges in modelling human gastric and small intestinal digestion, *Food Bioprocess Technol.* 6 (2012) 628–647; www.icbpharmapl/techno_slow.html.
- [20] M. Hoseinnejad, S.M.I. Katouzian, Inorganic and metal nanoparticles and their antimicrobial activity in food packaging applications, *Crit. Rev. Microbiol.* 44 (2018) 161–181.
- [21] H. Almasi, P.L. Mehryar, Fabrication of novel nanohybrids by impregnation of CuO nanoparticles into bacterial cellulose and chitosan nanofibers: Characterization, antimicrobial and release properties, *Carbohydr. Polym.* 186 (2018) 273–281.
- [22] T. Aruna, S. Mukasyan, Current Opinion in Solid State and Materials, *Science* 12 (2008) 44.
- [23] B. Qiu, X.F. Xu, R.H. Deng, G.Q. Xia, X.F. Shang, P.H. Zhou, Construction of chitosan / ZnO nanocomposite film by in situ precipitation, *Int. J. Biol. Macromol.* 122 (2019) 82–87, <https://doi.org/10.1016/j.ijbiomac.2018.10.084>.
- [24] M. Jayalakshmi, M. Palaniappan, K. Balasubramanian, Review of nanostructural aspects of metal electrodeposition, *International Journal of Electrochemical Science* 3 (2008) 96.
- [25] Y.X. Wen, H. Xiao, Y.L. Gan, H.F. Su, F. Wang, An overview on synthesis of metal oxide, *Journal of Inorganic Materials* 23 (2008) 286.
- [26] P. Dinka, A.S. Mukasyan, In situ preparation of oxide based supported catalysts by solution combustion synthesis, *J. of Physical Chemistry, B* 109 (2005) 21627.

- [27] N. Kasapoglu, A. Baykal, Y. Koseoglu, M.S. Toprak, Microwave assisted combustion synthesis of CoFe_2O_4 with urea, and its magnetic characterization, *Scr. Mater.* 57 (2007) 441.
- [28] J.H. Ryu, J.W. Yoon, K.B. Shim, Blue luminescence of nanocrystalline PbWO_4 phosphor synthesized via a citrate complex route assisted by microwave irradiation, *Solid State Commun.* 133 (2005) 657.
- [29] A. Mesaros, B.S. Vasile, D. Toloman, O.L. Pop, T. Marinca, M. Unguresan, I. Perhaita, M. Filip, F. Iordache, Towards understanding the enhancement of antibacterial activity in manganese doped ZnO nanoparticles, *Appl. Surface. Sci.* 471 (2019) 960–972, <https://doi.org/10.1016/j.apsusc.2018.12.086>.
- [30] M. Singhal, V. Chhabra, P. Kang, D.O. Shah, Synthesis of ZnO nanoparticles for varistor application using Zn-substituted aerosol OT microemulsion, *Mater. Res. Bull.* 32 (1997) 239.
- [31] V.L. Prasanna, R. Vijayaraghavan, Insight into the Mechanism of Antibacterial Activity of ZnO: Surface Defects Mediated Reactive Oxygen Species Even in the Dark, *Langmuir* 31 (33) (2015) 9155–9162, <https://doi.org/10.1021/acs.langmuir.5b02266>.
- [32] R. Zhong, Q. Zhong, M. Huo, B. Yang, H. Li, Preparation of biocompatible nano-ZnO/chitosan microspheres with multi-functions of antibacterial, UV-shielding and dye photodegradation, *International Journal of Biological Macromolecules* (2019), <https://doi.org/10.1016/j.ijbiomac.2019.09.217>.
- [33] C.F. João, K.T. Freni, C.S. José, S. Óscar, M.J. Ramos, E.P. Manuela, M.F. Xavier, Antimicrobial effects of chitosans and chitooligosaccharides, upon *Staphylococcus aureus* and *Escherichia coli*, in food model systems, *Food Microbiol.* 25 (2008) 922–928.
- [34] P. Petkova, A. Francesco, M.M. Fernandes, E. Mendoza, I. Perelshtein, A. Gedanken, T. Tzanov, Sonochemical coating of textiles with hybrid ZnO/chitosan antimicrobial nanoparticles, *ACS Appl. Mater. Interfaces* 6 (2014) 1164–1172.
- [35] N.V. Majeti, R. Kumar, A review of chitin and chitosan applications, *React. Funct. Polym.* 46 (2000) 1–27.
- [36] F. El-Kafrawy, S.M. El-Saeed, R.K. Farag, H.A. El-Saied, M.E.S. Abdel-Raouf, *Egypt. J. Pet.* 26 (23) (2017), <https://doi.org/10.1016/j.ejpe.2016.02.007>.
- [37] H. Al-Aidy, E. Amdeha, Green adsorbents based on polyacrylic acid-acrylamide grafted starch hydrogels: the new approach for enhanced adsorption of malachite green dye from aqueous solution, *International Journal of Environmental Analytical Chemistry* (2020), <https://doi.org/10.1080/03067319.2020.1711896>.
- [38] M. Rinaudo, Chitin and chitosan: properties and applications, *Prog. Polym. Sci.* 31 (2006) 603–632.
- [39] S.M. Seyedi, B. Anvaripour, M. Motavassel, N. Jadidi, Comparative cadmium adsorption from water by nanochitosan and Chitosan, *International Journal of Engineering and Innovative Technology (IJEIT)* 2 (9) (2013) 145–148.
- [40] A. Gamage, F. Shahidi, Use of chitosan for the removal of metal ion contaminants and proteins from water, *Food Chemistry* 104 (3) (2007) 989–996, <https://doi.org/10.1016/j.foodchem.2007.01.004>.
- [41] F.A. Ahing, N. Wid, Optimization of shrimp shell waste deacetylation for chitosan production, *International Journal of Advanced and Applied Sciences* 3 (10) (2016) 31–36, <https://doi.org/10.21833/ijaas.2016.10.006>.
- [42] I. Younes, M. Rinaudo, Chitin and chitosan preparation from marine sources; Structure, properties and applications, *Marine Drugs* 13 (3) (2015) 1133–1174, <https://doi.org/10.3390/md13031133>.
- [43] M.S. Benhabiles, R. Salah, H. Lounici, N. Drouiche, M.F.A. Goosen, N. Mameri, Antibacterial activity of chitin, chitosan and its oligomers prepared from shrimp shell waste, *Food Hydrocolloids* 29 (1) (2012) 48–56, <https://doi.org/10.1016/j.foodhyd.2012.02.013>.
- [44] S. Vaezifar, S. Razavi, M.A. Golozar, S. Karbasi, M. Morshed, M. Kamali, Effects of some parameters on particle size distribution of Chitosan nanoparticles prepared by ionic gelation method, *Journal of Cluster Science* 24 (3) (2013) 891–903.
- [45] M.E. Ali, Synthesis and adsorption properties of chitosan-CDTA-GO nanocomposite for removal of hexavalent chromium from aqueous solutions, *Arabian Journal of Chemistry* (2016), <https://doi.org/10.1016/j.arabjc.2016.09.010>.
- [46] L. Qi, Z. Xu, Lead sorption from aqueous solutions on chitosan nanoparticles, *Colloids and Surfaces A: Physicochemical and Engineering Aspects* 251 (1–3) (2004) 183–190, <https://doi.org/10.1016/j.colsurfa.2004.10.010>.
- [47] C. Liewhiran, S. Seraphin, S. Phanichphant, Synthesis of nano-sized ZnO powders by thermal decomposition of zinc acetate using *Broussonetiapyrifera* (L.) Vent pulp as a dispersant, *Curr. Appl. Phys.* 6 (2006) 499–502.
- [48] Y.A. Soliman, A.M. Ibrahim, A.H. Moustafa, M.A. Hamed, Antifouling evaluation of extracts from Red sea soft corals against primary biofilm and biofouling, *Asian Pac. J. Trop Biomed.* 7 (11) (2017) 991–997.
- [49] Y.A. Soliman, A.M. Ibrahim, H. r. z. Tadmor, A.E. Abou Taleb, A.H. Moustafa, M.A. Hamed, Antifouling and antibacterial activities of marine bioactive compounds extracted from some Red sea cucumber, *International J. of Contemporary Applied sciences* 3 (9) (2016) 83–103.
- [50] M.S. Hossain, A. Iqbal, Production and characterization of chitosan from shrimp waste, *J. Bangladesh Agricultural University* 12 (1) (2014) 153–160.
- [51] A.M. Alsharabasy, Semi-synthesis of chitosan with high molecular weight and enhanced deacetylation degree, *Polymer Science* 2 (2016) 2.
- [52] Q. Zhong, J. Tian, T. Liu, Z. Guo, S. Ding, H. Li, Preparation and antibacterial properties of carboxymethyl chitosan/ZnO nanocomposite, *Materials letters* 212 (2018) 58–61, <https://doi.org/10.1016/j.matlet.2017.10.062>.
- [53] D.R. Bhumkar, V.B. Pokharkar, Studies on Effect of pH on Cross-linking of Chitosan with Sodium Tripolyphosphate: A Technical Note, *AAPS Pharm SciTech* 7 (2) (2006) Article 50.
- [54] S.D. Sarkar, B.L. Farrugia, T.R. Dargavill, S. Dhara, Physico-Chemical/Biological Properties of Tripolyphosphate Cross-Linked Chitosan Based Nanofibers, *Materials Science and Engineering C* 33 (2013) 1446–1454.
- [55] Z. Yang, Q.H. Liu, The structural and optical properties of ZnO nanorods via citric acid-assisted annealing route, *Journal of Materials Science* 43 (19) (2008) 6527–6530.
- [56] G. Huang, L.Q. Mo, Y.X. Wei, H. Zhou, Y.A. Guo, S.J. Wei, Effect of Mesoporous Chitosan Action and Coordination on the Catalytic Activity of Mesoporous Chitosan-Grafted Cobalt Tetrakis(p-Sulfophenyl)Porphyrin for Ethylbenzene Oxidation, *Catalysts* 8 (2018) 199.
- [57] N.R. Panda, B.S. Acharya, P. Nayak, B.P. Bag, *Ultrason. Sonochem* 21 (2014) 582–589.
- [58] M.S. Sivakami, T. Gomathi, J. Venkatesan, H.S. Jeong, S.K. Kim, P.N. Sudha, Preparation and characterization of nano chitosan for treatment wastewaters, *International Journal of Biological Macromolecules* 57 (2013) 204–212.
- [59] E. Tang, G. Cheng, X. Ma, X. Pang, Q. Zhao, Surface modification of ZnO nanoparticles by PMAA and its dispersion in aqueous system, *Appl. Surf. Sci.* 252 (14) (2006) 5227–5232.